

Global Health, Economy & Governance Dynamics: Project 2

Track A: Tableau (Advanced)

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1. Introduction and Background

In Project 1, we examined the relationship between economic growth, democratic governance, and health outcomes across 236 countries from 1960 to 2024. Using two joined datasets, a combined Gapminder/World Bank dataset (15,340 rows, 15 variables) and the V-Dem v16 governance dataset (~11,000 rows, 6 variables), we produced six visualizations and a dashboard exploring whether wealth consistently translates to longer lives, whether democratic institutions improve health, and whether world regions are converging or diverging in development. Our key finding was that while economic growth broadly correlates with improved life expectancy, the relationship is neither linear nor universal: governance quality, regional context, and crisis events all mediate the connection between wealth and health.

Project 2 deepens this analysis by pursuing three questions that Project 1 raised but could not answer. First, our Project 1 heatmap revealed a suggestive pattern between corruption and health outcomes, but only for 30 cherry-picked countries; it never tested whether that pattern holds across all income levels. Second, although our dataset includes a temporal income classification variable (Income Class WB, 1987–2024), Project 1 treated income groups as static snapshots, never visualizing how countries move between income classes over time. Third, our Project 1 line chart showed a visible COVID-era dip in life expectancy but did not quantify the drop, compare it across income groups, or examine which countries recovered. Project 2 addresses each of these gaps with new research questions, new chart types, and **advanced Tableau features not used in Project 1**.

We chose to continue with Track A (Tableau) because the advanced features available (parameter controls, table calculations, Pages shelf animation, and INCLUDE LOD expressions) allowed us to build interactive, exploratory visualizations that empower the viewer to test hypotheses directly, an analytical depth that static charts cannot provide. **New Research Questions:**

- **Q6:** Does corruption kill? Do less corrupt countries achieve better health outcomes at every income level?
- **Q7:** Which countries climbed the income ladder, and how has income class mobility related to health gains since 1987?
- **Q8:** How did COVID-19 impact different regions and income groups? Which countries were hit hardest and which recovered fastest?

2. Datasets

We used the same two datasets from Project 1, already prepped in Tableau Prep, with no additional structural changes. Dataset A (Gapminder + World Bank) provides country-level life expectancy, GDP per capita, population, and income classifications for 236 countries (1960–2024). Dataset B (V-Dem v16) provides indices on electoral democracy, liberal democracy, and political corruption for 182 countries. The datasets are connected via a Tableau relationship on *iso_code* and *year*.

For Project 2, we made greater use of two variables that were underutilized in Project 1. The *income_class_wb* variable, which records the World Bank’s annual income classification for each country from 1987 to 2024, became the basis for our income mobility analysis (Q7). The *v2x_corr* variable (political corruption index, scaled 0–1 where higher values indicate greater corruption) was previously used only in a static heatmap of 30 countries; we now apply it across all countries and income groups (Q6).

We also created several new calculated fields for Project 2. **FIXED LOD** expressions were used to compute *Life Exp 2019*, *Life Exp 2020*, and *Life Exp 2022* for each country, enabling calculations of COVID impact and recovery. A *Corruption Level v2* field classifies each country-year observation as “High Corruption” or “Low Corruption” based on a user-adjustable **parameter** threshold. A *COVID Drop* field computes the difference between 2020 and 2019 life expectancy, while *COVID Recovery* captures the rebound from 2020 to 2022.

One limitation that remains from Project 1 is that the income classification variable is available only from 1987 onward, restricting our income mobility analysis to a 37-year window. Additionally, V-Dem corruption data is unavailable for some smaller nations, resulting in null values that we filtered out of our analysis.

3. Visual Analysis and Findings

3.1 COVID Impact: Year-over-Year Life Expectancy Change (Map)

Mapping the pandemic’s toll on life expectancy across nations (Q8)

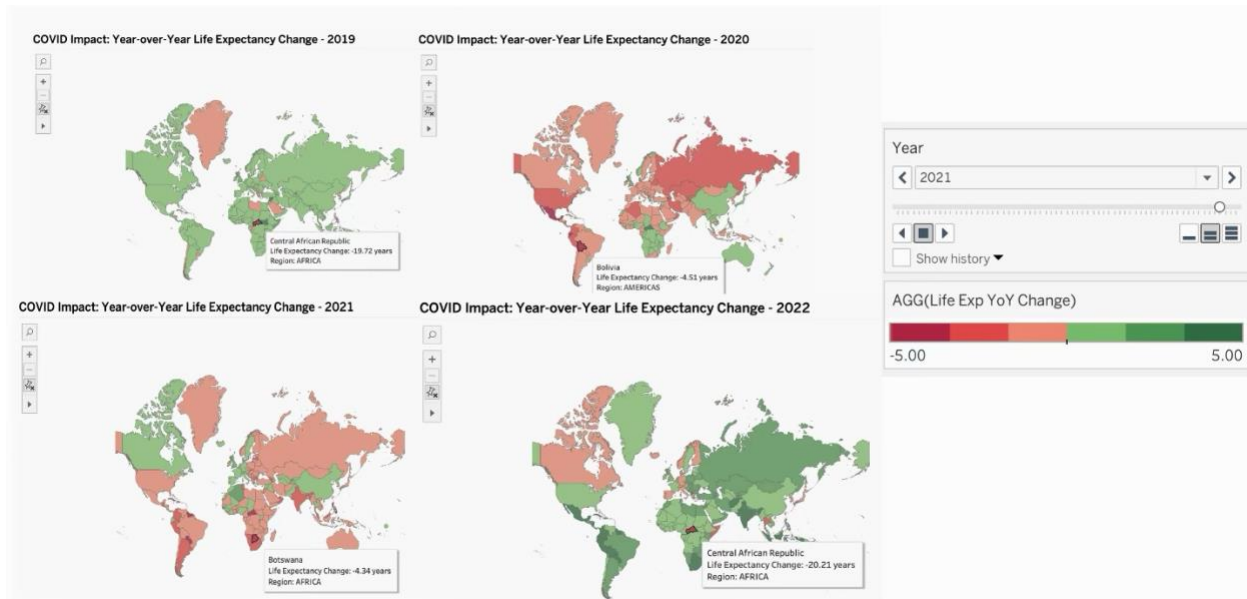
Our first Project 2 visualization uses a choropleth map with a table calculation to show year-over-year change in life expectancy for every country. The measure is computed as $AVG([Life Expectancy]) - LOOKUP(AVG([Life Expectancy]), -1)$, a table calculation that computes the difference between each year and the previous year, partitioned by country. We applied a red-green diverging color palette centered on zero, with the range fixed at -5 to +5 years, so that red indicates declining life expectancy and green indicates improving life expectancy. The Year dimension is placed on the **Pages shelf**, enabling animated playback, one of our advanced features not used in Project 1.

The map reveals a striking pattern when animated from 2019 to 2020: nearly the entire world shifts from green or neutral to deep red, with Latin America, Europe, and Central Asia showing the most

severe declines. By 2021, parts of Asia and Africa begin recovering (turning green), while the Americas and Europe remain red. By 2022, a global recovery is underway, with most countries returning to pre-pandemic trajectories. This animation makes the pandemic’s geographic and temporal spread viscerally clear in a way that a static chart cannot, following Schneiderman’s (1996) visual information-seeking mantra of “overview first, zoom and filter, then details on demand.” The Pages shelf animation provides an overview; clicking individual countries displays details via tooltips.

COVID Impact: Year-over-Year Life Expectancy Change (2019–2022)

The pandemic’s geographic spread and recovery, animated via the Pages shelf



3.2 Does Corruption Kill? Life Expectancy by Income Group and Corruption Level

Exploring the corruption-health relationship across income levels (Q6)

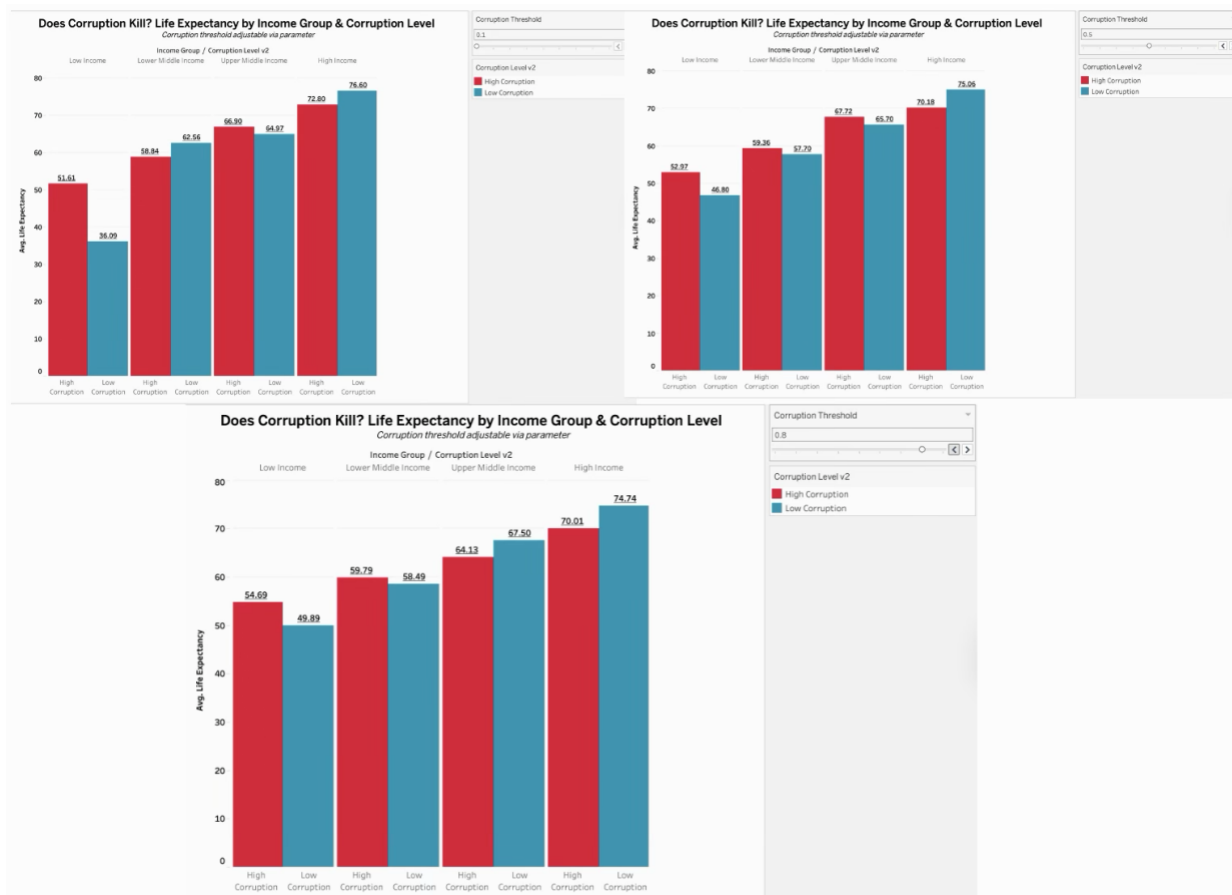
This grouped bar chart directly tests Q6 by comparing average life expectancy between high-corruption and low-corruption countries within each World Bank income group. The key innovation is **parameter control**; a slider that allows the viewer to adjust the corruption threshold (ranging from 0.1 to 0.9), which defines the boundary between “high” and “low” corruption. This interactivity transforms the chart from a single static assertion into a hypothesis-testing tool, which we consider the most analytically powerful feature in our Project 2 work.

The *Corruption Level* v2 calculated field classifies each observation as *IF [V2X Corr] <= [Corruption Threshold] THEN "Low Corruption" ELSE "High Corruption" END*, operating at the row level rather than on aggregated values. Income Group is sorted logically from Low Income to High Income on the x-axis, with paired red (High Corruption) and teal (Low Corruption) bars within each group.

The findings are nuanced and challenge a simplistic “corruption kills” narrative. At a strict threshold of 0.1, the cleanest governments do produce better health outcomes across all income groups. However, at the midpoint threshold of 0.5, the pattern partially reverses: highly corrupt countries in the Low Income and Lower Middle-Income groups actually show higher average life expectancy than their low-corruption peers, while the expected pattern holds only for High Income countries. At a lenient threshold of 0.8, high-corruption countries lead in most categories. This suggests that corruption’s health penalty is most evident at high income levels, whereas in poorer countries, other factors, perhaps foreign aid, geography, disease burden, or resource wealth, may dominate health outcomes regardless of governance quality. As Cleveland and McGill (1984) demonstrated, grouped bar charts exploit the pre-attentive feature of position along a common scale, making these within-group comparisons perceptually accurate and immediately readable.

Life Expectancy by Income Group at Varying Corruption Thresholds (0.1, 0.5, 0.8)

Parameter control allows the viewer to test the robustness of the corruption-health relationship



3.3 Health Outcomes by Corruption Level Across Income Groups

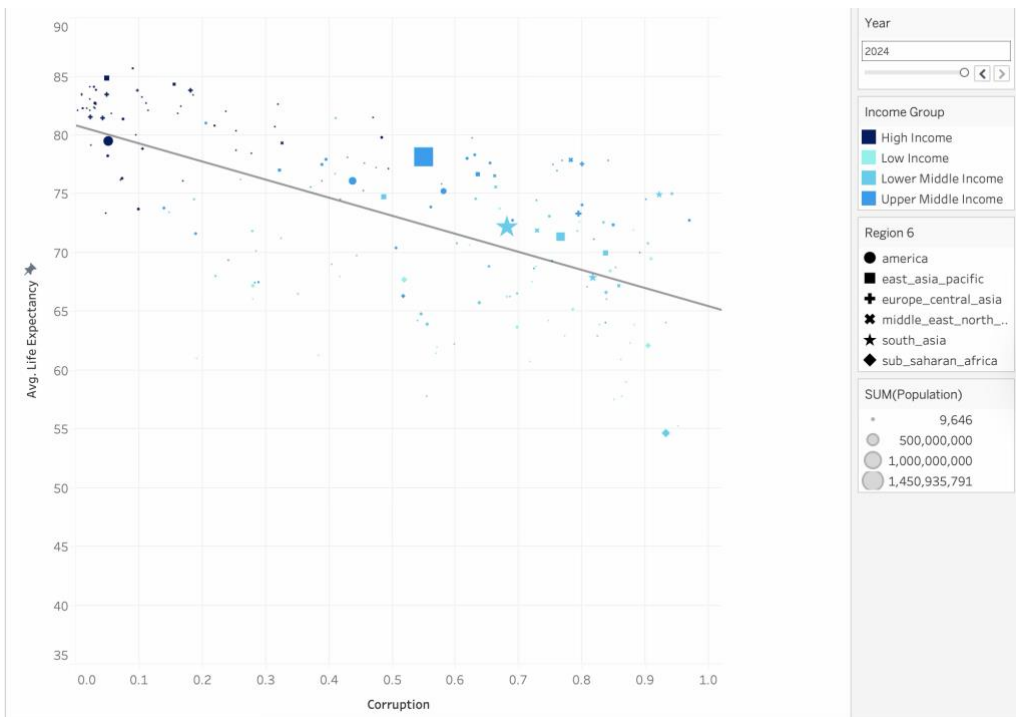
Country-level corruption and life expectancy with trend analysis (Q6)

To complement the grouped bar chart's aggregated view, this scatter plot maps individual countries with corruption (V2X Corr, 0–1) on the x-axis and average life expectancy on the y-axis. Each point represents a country, sized by population and shaped by Region 6, with color encoding Income Group. A trend line reveals the overall negative relationship: as corruption increases, life expectancy tends to decline. A Year parameter allows the viewer to examine this relationship at different points in time; the 2024 snapshot is shown here as the most policy-relevant view.

This chart addresses a limitation of the bar chart by showing the full distribution rather than just group averages. The scatter reveals considerable variance within each income group. Some highly corrupt countries achieve relatively strong health outcomes, while some low-corruption nations underperform. The cluster of High-Income countries (dark blue) in the upper-left corner confirms that wealth and good governance tend to co-occur, with most achieving life expectancy above 80 years at corruption levels below 0.2. Meanwhile, Upper Middle-Income countries with moderate corruption (0.4–0.6) still achieve life expectancy around 75–78 years, supporting the bar chart’s finding that corruption’s health penalty is most pronounced at higher income levels. The lone Sub-Saharan African diamond near the bottom right, high corruption, and a life expectancy of around 55 illustrate the compounding disadvantage of poverty and poor governance. As Franconeri et al. (2021) note, scatter plots are effective for revealing correlations, outliers, and clusters simultaneously, making them an ideal complement to the categorical comparison of the bar chart.

Corruption and Life Expectancy: Country-Level Patterns Across Income Groups (2024)

Higher corruption correlates with lower life expectancy, but income group and region mediate the relationship



3.4 The COVID Dip: How Much Did Life Expectancy Fall? And Who Recovered?

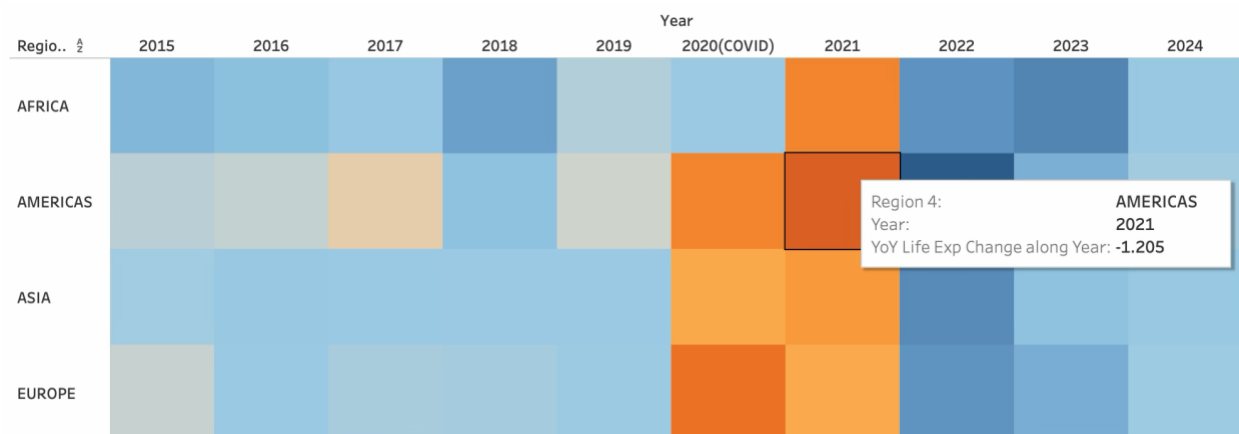
Regional patterns of pandemic impact and recovery (Q8)

This heatmap displays year-over-year changes in life expectancy by region (rows) and year (columns), using an orange-to-blue diverging color palette, where orange indicates decline and blue indicates improvement. The visualization covers 2015–2024, providing context both before and after the pandemic. The year-over-year change is computed using the same table calculation as the map ($AVG([Life\ Expectancy]) - LOOKUP(AVG([Life\ Expectancy]), -1)$), aggregated at the regional level.

The heatmap makes three patterns immediately visible. First, the pre-COVID period (2015–2019) shows predominantly blue tones across all regions, reflecting steady global improvement in life expectancy. Second, 2020 marks a dramatic shift to orange, with Europe and the Americas showing the deepest orange, the heaviest immediate toll, while Asia experienced a lighter decline. Third, the sustained and recovery patterns diverged sharply: in 2021, the Americas remained the darkest orange, followed by Africa, indicating a prolonged pandemic impact, while Europe had already begun to recover. By 2022, the Americas show the strongest rebound (darkest blue), followed by Asia and Europe, with Africa lagging behind. Africa's strongest recovery did not appear until 2023, when it shows the darkest blue, while the Americas and Asia had already settled into lighter tones. This chart complements the country-level map and rankings table by providing the regional, temporal context that neither can show alone. Few (2006) recommends heatmaps for revealing patterns across two categorical dimensions, and the color intensity makes cross-regional comparisons immediate without requiring the viewer to trace lines or compare bar heights.

The COVID Dip: Year-over-Year Life Expectancy Change by Region (2015–2024)

All regions plunged in 2020, but recovery speed varied — Asia and Africa rebounded fastest

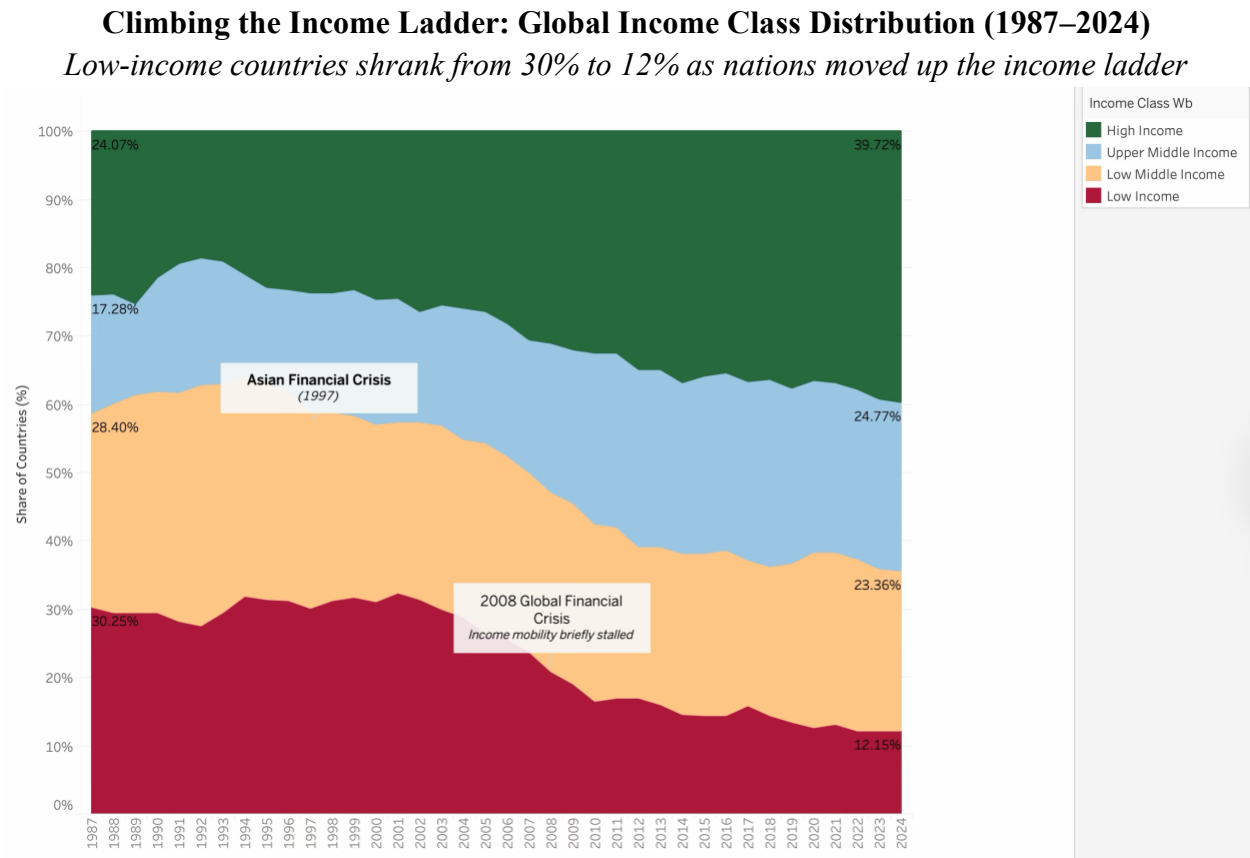


3.5 Climbing the Income Ladder: Global Income Class Distribution, 1987–2024

How has income mobility reshaped the world? (Q7)

This stacked area chart shows the share of countries in each World Bank income class over time, using a **Percent of Total** table calculation, our second advanced table calculation feature for Project 2. The x-axis spans 1987 to 2024 (the full availability window of the *income_class_wb* variable), and the y-axis displays the percentage of countries falling in each income class. The stacking order places Low Income at the bottom and High Income at the top, so that upward income mobility is visually represented as upward movement on the chart.

The visualization tells a compelling story of global economic progress. In 1987, low-income countries comprised approximately 30% of all classified nations; by 2024, that share had fallen to roughly 12%. Simultaneously, the High-Income share grew from 24% to nearly 40%. The Low, Middle, and Upper Middle-Income layers exhibit more complex dynamics, with the Upper Middle-Income layer expanding notably. Two annotations highlight moments when this upward trajectory was disrupted: the 1997 Asian Financial Crisis and the 2008 Global Financial Crisis, during which income mobility briefly stalled as some countries slipped back a category. The area chart follows Munzner’s (2014) principle of using area marks to encode part-to-whole relationships over time, making compositional change immediately apparent.



3.6 COVID-19 Impact Rankings: Hardest Hit Countries

Which countries suffered the steepest life expectancy declines in 2020? (Q8)

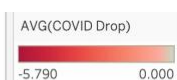
The final Project 2 chart is a ranked text table showing the 20 countries with the largest life expectancy drop from 2019 to 2020. Columns include Rank, Country Name, Region, Income Group, COVID Drop (the 2019-to-2020 difference), Recovery (the 2020-to-2022 rebound), and life expectancy values for 2019, 2020, and 2022. The ranking uses an **INDEX** table calculation, and the COVID Drop field uses **FIXED LOD** expressions to anchor each country's 2019 and 2020 life expectancy values, regardless of dimensional filters. Cell-level heatmap shading applies a red gradient to the COVID Drop column, with darker red indicating more severe declines.

The table reveals that the hardest-hit countries are not the poorest. Armenia leads with a 5.79-year drop, followed by Ecuador (-5.05), Bolivia (-4.51), and Andorra (-4.31). The top 20 is dominated by the Americas (7 countries) and Europe (6 countries), with Upper Middle-Income countries disproportionately represented. Only one low-income country (Tajikistan, ranked 14th) appears. This counterintuitive finding — that wealthier countries with older, more urbanized populations experienced sharper COVID-era life expectancy declines than poorer nations — adds important nuance to Q8 and connects back to the regional heatmap's finding that the Americas and Europe were hardest hit. The recovery column shows that most countries rebounded strongly by 2022, with Armenia recovering in 5.28 years and Ecuador in 4.36 years, though some countries, like Oman (0.04), show minimal recovery. As Tufte (2001) argued, well-designed tables can convey precise quantitative information more efficiently than charts when the audience needs to compare specific values across multiple dimensions.

COVID-19 Impact Rankings: 20 Hardest Hit by Life Expectancy Drop (2019–2020)

Americas & Europe dominate, with Upper-Middle Income countries disproportionately affected

Rank of Avg.	Country Name	Region 4	Income Group	COVID Drop	Recovery	LE 2019	LE 2020	LE 2022
1	Armenia	EUROPE	Upper Middle Income	-5.79	5.28	74.93	69.14	74.41
2	Ecuador	AMERICAS	Upper Middle Income	-5.05	4.36	77.29	72.24	76.61
3	Bolivia	AMERICAS	Lower Middle Income	-4.51	4.13	67.90	63.39	67.52
4	Andorra	EUROPE	High Income	-4.31	4.25	84.14	79.83	84.08
5	Mexico	AMERICAS	Upper Middle Income	-4.01	3.44	74.50	70.50	73.94
6	Kosovo	EUROPE	Upper Middle Income	-2.96	3.33	77.21	74.25	77.58
7	Azerbaijan	EUROPE	Upper Middle Income	-2.90	3.75	73.23	70.32	74.07
8	Nicaragua	AMERICAS	Lower Middle Income	-2.90	3.56	73.72	70.82	74.38
9	Iran	ASIA	Upper Middle Income	-2.68	2.63	76.91	74.23	76.86
10	San Marino	EUROPE	High Income	-2.50	2.96	85.19	82.69	85.65
11	Algeria	AFRICA	Upper Middle Income	-2.38	2.83	75.72	73.34	76.17
12	Kazakhstan	ASIA	Upper Middle Income	-2.33	2.23	73.47	71.14	73.37
13	Qatar	ASIA	High Income	-2.33	1.18	83.12	80.79	81.97
14	Tajikistan	ASIA	Low Income	-2.30	2.90	70.96	68.65	71.55
15	North Macedonia	EUROPE	Upper Middle Income	-2.29	2.15	76.79	74.49	76.65
16	Peru	AMERICAS	Upper Middle Income	-2.15	2.66	76.34	74.19	76.85
17	Oman	ASIA	High Income	-2.15	0.04	80.15	78.00	78.04
18	Panama	AMERICAS	High Income	-2.08	2.88	78.55	76.46	79.34
19	Colombia	AMERICAS	Upper Middle Income	-1.94	1.66	76.80	74.86	76.52
20	Lebanon	ASIA	Upper Middle Income	-1.87	1.66	78.13	76.26	77.92



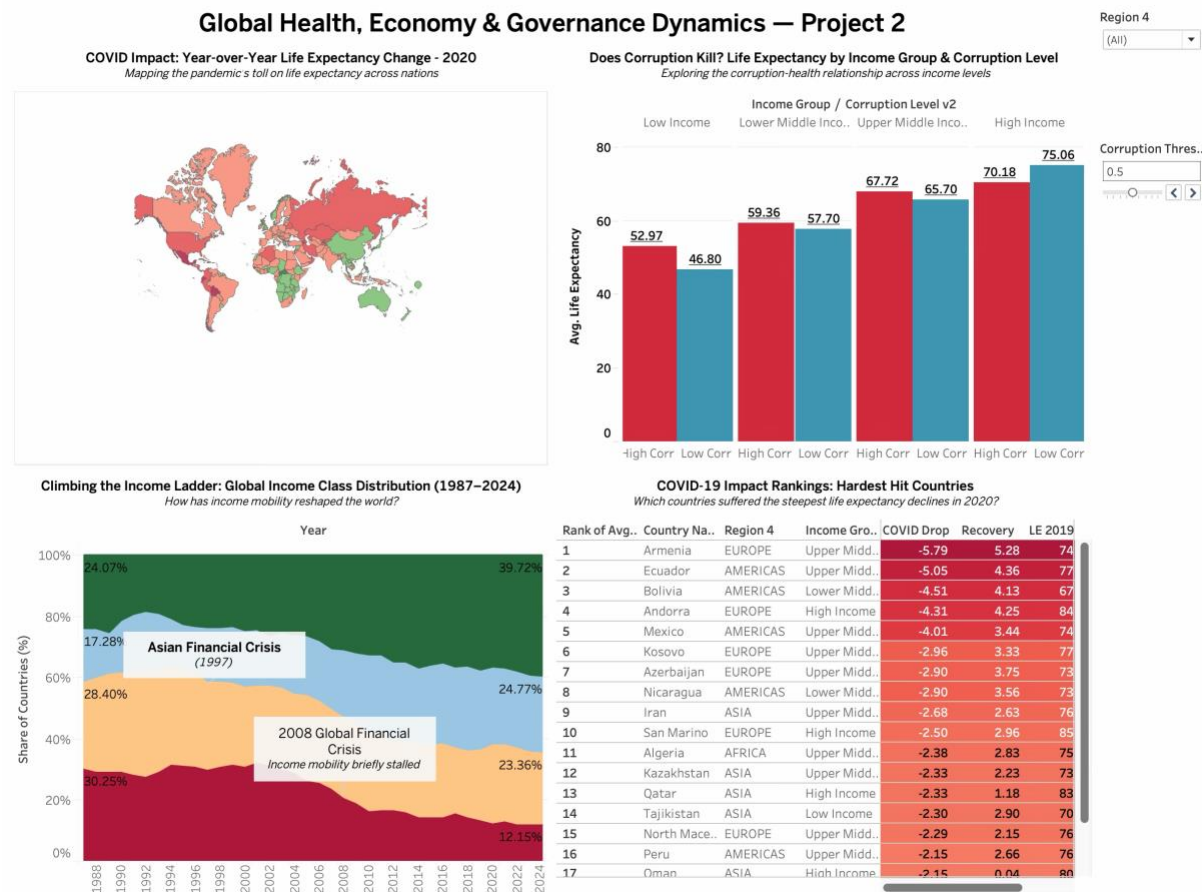
3.7 Project 2 Dashboard

Integrating four views with interactive filtering (Q6, Q7, Q8)

The Project 2 dashboard arranges four charts: the **COVID map**, **corruption bar chart**, **income mobility area chart**, and **COVID rankings table**, in a 2×2 grid following Few’s (2006) principle that dashboards should present the most important information on a single screen without requiring scrolling. Interactive elements include a **Region 4 dropdown filter** (applying to all charts simultaneously), a Corruption Threshold **parameter slider** (controlling the bar chart’s corruption classification), and the map functioning as a **cross-filter** (clicking a country filters the other three views). These interactive elements embody Shneiderman’s (1996) mantra by enabling the viewer to move from a global overview to filtered, detailed analysis without leaving the dashboard.

Compared to our Project 1 dashboard, Project 2 improves in three ways. First, parameter control adds genuine interactivity, allowing viewers to test their own hypotheses rather than passively receiving our conclusions. Second, the cross-filtering between the map and table creates a linked analysis workflow; selecting a region on the map instantly highlights that region’s COVID rankings. Third, the layout is more balanced, with each chart occupying an equal quadrant and no visual hierarchy issues from oversized legends or controls.

Project 2 Dashboard: COVID Impact, Corruption, and Income Mobility
Interactive exploration via region filter, Corruption Threshold parameter, & map cross-filtering



4. Summary and Conclusions

Project 2 extended our Project 1 analysis in three substantive directions. Our corruption analysis (Q6) revealed that the relationship between governance quality and health outcomes is more complex than a simple “corruption kills” narrative: the pattern depends heavily on income level and on how strictly corruption is defined, a nuance made explorable through the parameter control. Our income mobility analysis (Q7) documented significant global progress, with the share of low-income countries falling from 30% to 12% over 37 years, while also identifying moments when financial crises temporarily reversed these gains. Our COVID analysis (Q8) quantified the pandemic’s toll on life expectancy and uncovered the counterintuitive finding that wealthier, more urbanized countries experienced sharper declines than poorer nations.

The advanced Tableau features used in Project 2: table calculations (year-over-year change and percent of total), parameter controls (corruption threshold slider), Pages shelf animation (COVID map playback), and FIXED LOD expressions (COVID impact calculations); each deepened the analytical story in ways that Project 1’s static visualizations could not achieve. The parameter control, in particular, transformed our corruption analysis from a single finding into an interactive exploration tool. Several data limitations from Project 1 remain. The income classification variable is available only from 1987, limiting our income mobility analysis. V-Dem corruption data is missing for some countries, potentially biasing our corruption analysis toward better-documented nations. Life expectancy data for 2023–2024 may be provisional in some countries, and the COVID-era drops could reflect both direct mortality and indirect effects (disrupted healthcare, delayed treatment) that our data cannot distinguish.

If this project continued, we would explore sub-national variation in COVID impact (urban vs. rural), test whether democratic institutions moderated the pandemic's health toll, and examine whether countries that climbed the income ladder also improved their governance scores — linking all three of our Project 2 questions into a unified analytical framework.

5. Contributions

Team Member	Contribution
Tambudzai Charumbira	COVID Impact Map, Corruption Bar Chart, Income Mobility Area Chart, COVID Rankings Table, Dashboard design, Written narrative
Sayan Patra	Corruption Scatter Plot, COVID Dip Heatmap, Data preparation, 2 nd Dashboard design, Presentation script

5. References

- Cleveland, W. S., & McGill, R. (1984). Graphical Perception: Theory, Experimentation, and Application to the Development of Graphical Methods. *Journal of the American Statistical Association*, 79(387), 531–554. <https://doi.org/10.1080/01621459.1984.10478080>
- Coppedge, M., Gerring, J., Knutsen, C. H., Lindberg, S. I., Teorell, J., Altman, D., Bernhard, M., Fish, M. S., Glynn, A., & Hicken, A. (2025). V-Dem Codebook v16. Varieties of Democracy (V-Dem) Project. Varieties of Democracy. <https://www.v-dem.net/>
- Few, S. (2006). *Information dashboard design: The effective visual communication of data*. O'Reilly Media.
- Franconeri, S. L., Padilla, L. M., Shah, P., Zacks, J. M., & Hullman, J. (2021). The Science of Visual Data Communication: What Works. *Psychological Science in the Public Interest*, 22(3), 110–161. <https://doi.org/10.1177/15291006211051956>
- Gapminder Foundation. (2024). Data documentation. <https://www.gapminder.org/data/documentation/>
- Munzner, T. (2014). *Visualization Analysis and Design*. A K Peters/CRC Press. <https://doi.org/10.1201/b17511>
- SHNEIDERMAN, B. (1996). The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations. *Proceedings of the IEEE Symposium on Visual Languages (VL'96)*. <https://cir.nii.ac.jp/crid/1573950399986655744>
- Tufte, E. R. (2001). *The visual display of quantitative information* (2nd ed.). Graphics Press.
- World Bank. (2024). World Bank Country and Lending Groups – World Bank Data Help Desk. <https://datahelpdesk.worldbank.org/knowledgebase/articles/906519>
- World Health Organization. (2022). Global health estimates Life expectancy and healthy life expectancy. <https://www.who.int/data/gho/data/themes/mortality-and-global-health-estimates/ghe-life-expectancy-and-healthy-life-expectancy>